

Spread Locatus

Q1). What is Statistical Distributions?

→ A statistical distribution describes how data values are spread across possible outcomes. It shows the probability of occurrence of different values.

- Helps in understanding patterns in data
- Used in prediction and modeling
- Examples : Normal, Binomial, Poisson

Q2). What is Q-Q Plot and why it is used?

→ A Q-Q (Quantile - Quantile) plot compares the distribution of a dataset with a theoretical distribution (usually normal).

Purpose :

- To check if data follows a normal distribution
- If points lie on straight line → data is normal
- If not → data is skewed or non-normal



Q3) Difference btw Discrete and Continuous Distributions

-) Features Discrete Continuous

i) Values Countable Infinite

ii) Eg. No. of transactions Transaction amount

iii) Graph Bar chart Smooth curve

Q4) What is ^{Binomial} Bernoulli Distribution

-) A Binomial Distribution represents multiple Bernoulli trials

- Fixed number of trials (n)
- Probability of success (p)

Eg: No. of successful transactions in week

Q5) What is Bernoulli Distribution

-) A Bernoulli distribution represents a single experiment with only two outcomes:

- Success (1)
- Failure (0)

Eg: Transaction success or failure

Q6). Explain Log-Normal Distribution

- A variable follows a log-normal distribution if its logarithm is normally distributed
- Used for skewed data
- Common in financial data (like transaction amounts)

Q7). Explain Power Law Distribution

- Power Law Distribution shows that
- Few values are very high
- Most values are small

Eg: Few customers make very high transactions.

Q8). What is Box-Cox Transform?

- Box-Cox is a transformation technique used to:
- Make data more normal
- Reduce skewness
- Stabilize variance

Q9) Explain Poisson Distribution with an example

→ Poisson distribution models the number of events in a fixed interval

Eg: Number of transaction per day

Q10) What is Z-score Probability?

→ Z-score measures how far a value is from the mean

$$Z = \frac{X - \mu}{\sigma}$$

used to

- Detect outliers
- Calculate probabilities

11) DF b/w PDF and CDF

PDF

CDF

i) It means Probability Density

i) It means Cumulative Probability

ii) It output is likelihood

ii) It output total probability till a point

Range is 0 to ∞

iii) Range is 0 to 1